

JEE MAIN REVISION MASTER: DEFINITE INTEGRATION

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 31 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	King's Property & Definite Integral Properties	30 PYQs	Integral Properties
Standard	Limit of a Sum & Area Under Curves	1 PYQs	Summation & Area

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
King's Property & Definite Integral Properties	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, $\int_{-a}^a f(x) dx = 0$ (if odd), $\frac{d}{dx} \int u(x)v(x) f(t) dt = f(v(x))v'(x) - f(u(x))u'(x)$
Limit of a Sum & Area Under Curves	$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = \int_0^1 f(x) dx$, $\text{Area between } y^2 = 4ax \text{ and } y = mx \text{ implies } \frac{8a^2}{3m^3}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 King's Property & Definite Integral Properties (Very High Yield — 30 PYQs)

Core Principle: King's property $f(a+b-x)$, reflection symmetry, integrals of even/odd functions, periodic functions, and Leibnitz integral differentiation.

■ **Caution / JEE Pitfall:** King's property added to the original integral often yields $2I = \int_a^b 1 dx = b - a \implies I = (b - a) / 2$.

■ **Speed Trick:** Integral of odd function over symmetric limits $[-a, a]$ is always ZERO: $\int_{-a}^a f(x) dx = 0$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int_{-6}^6 (a-6)^b - 6f(x+6) dx$ is equal to..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "One of the possible functions $f(x)$ which satisfies $\int_{-2}^2 f(x) dx = 0$ is..." [Ans: A]
- [2025 Shift] Q#1 (MCQ): " $\int_0^1 \log((1/x)-1) dx$ is..." [Ans: N/A]

3.2 Limit of a Sum & Area Under Curves (Standard Yield — 1 PYQs)

Core Principle: Definite integrals as limits of Riemann sums (r/n o x , $1/n$ o dx), and enclosed area calculations between curves.

■ **Caution / JEE Pitfall:** Area is always a positive geometric quantity; if the curve dips below the x-axis, take absolute value $|y|dx$.

■ **Speed Trick:** Area bounded by $y^2 = 4ax$ and $x^2 = 4by$ is given directly by $\frac{16ab}{3}$.

Representative Source PYQs:

- [2022 Shift] Q#2 (MCQ): "Evaluate $\int_1^2 (2)^{3x^2} dx$ as the limit of a sum..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$\int (a-6)^b - 6f(x+6)dx$ is equal to

- (A) $\int (a)^{bf(x+6)} dx$
- (B) $\int (a)^{bf(x-6)} dx$
- (C) $\int (a)^{bf(x)} dx$ **[Correct]**
- (D) $-\int (a)^{bf(x)} dx$

Correct Answer: C

Solution Analysis: Put $t=x+6$ Then $dt=dx$ Now change the limits. When $x=a-6$, we get $t=(a-6)+6=a$ When $x=b-6$, we get $t=(b-6)+6=b$ So, $\int_{(a-6)}^{(b-6)} b-6f(x+6)dx = \int_a^b b-6f(t)dt$ The letter used for the variable does not matter in a definite integral. So, $\int_a^b b-6f(t)dt = \int_a^b b-6f(x)dx$ Hence, $\int_{(a-6)}^{(b-6)} b-6f(x+6)dx = \int_a^b b-6f(x)dx$ So the correct option is Option C

Benchmark Problem 2: KCET 2022 (Q#2)

Evaluate $\int (2)^{3x^2} dx$ as the limit of a sum

- (A) $(72 / 6)$
- (B) $(53 / 9)$
- (C) $(25 / 7)$
- (D) $(19 / 3)$

Correct Answer: None

Solution Analysis: Let $I = \int_0^2 x^3 dx$, $J = \int_0^2 x^3 f(x) dx$, where $f(x) = x^2$

$$\begin{aligned} & \text{\texttt{\textbf{Also,}}} a=2, b=3 \\ & \lim_{n \rightarrow \infty} [f(2) + f(2+\frac{1}{n}) + f(2+\frac{2}{n}) + \dots + f(2+(n-1)\frac{1}{n})] \\ & \lim_{n \rightarrow \infty} [\frac{1}{n}(2^2 + (2+\frac{1}{n})^2 + (2+\frac{2}{n})^2 + \dots + (2+(n-1)\frac{1}{n})^2)] \\ & \lim_{n \rightarrow \infty} [\frac{1}{n}(2^2 + (2^2 + \frac{1}{n} + 4\frac{1}{n^2}) + (2^2 + 2\frac{1}{n} + 8\frac{1}{n^2}) + \dots + (2^2 + (n-1)^2\frac{1}{n^2} + 4(n-1)\frac{1}{n}))] \\ & \lim_{n \rightarrow \infty} [\frac{1}{n}(2^2 + \frac{1}{n} + 4\frac{1}{n^2} + 2^2 + 2\frac{1}{n} + 8\frac{1}{n^2} + \dots + (n-1)^2 + 4(n-1))] \end{aligned}$$